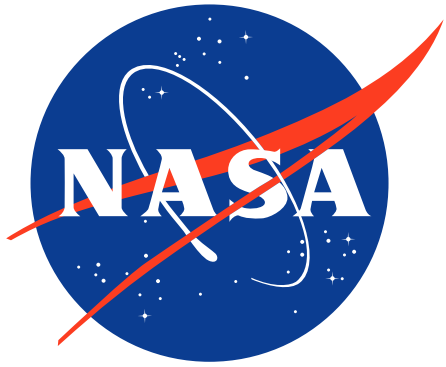




The Noise Measured by Back-illuminated Silicon Sensors

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IRIS Noise Mystery

- IRIS measures less noise than predicted (Wulser et al. 2018).
 - Photon transfer curve is too shallow
 - Expected slope ratio > 2
 - Measured slope ratio = 1.5
 - SNR is ~25% better than expected

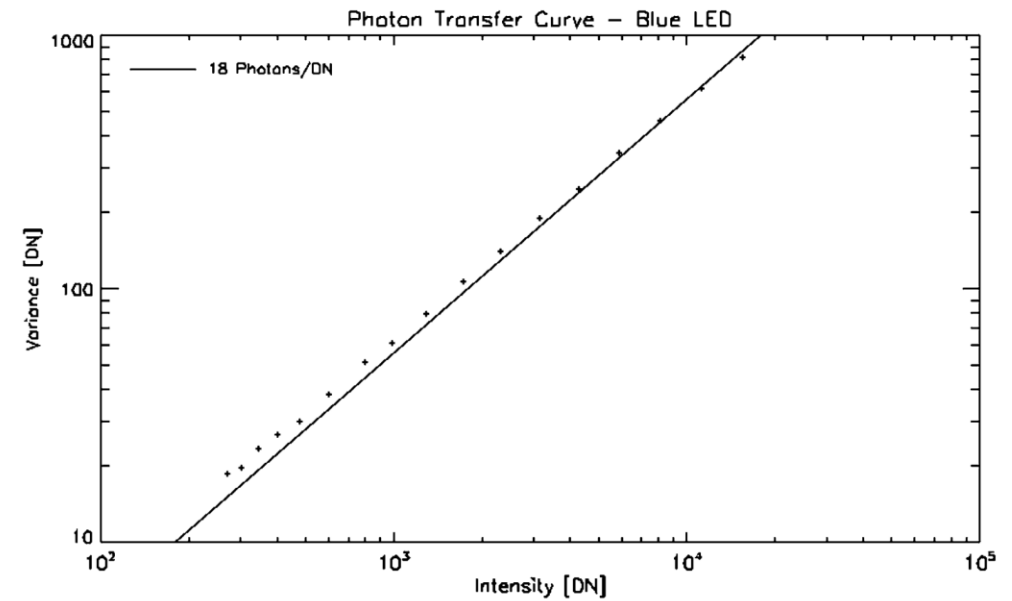


Figure 26 Photon transfer curve for the SJI CCD measure with a blue LED.

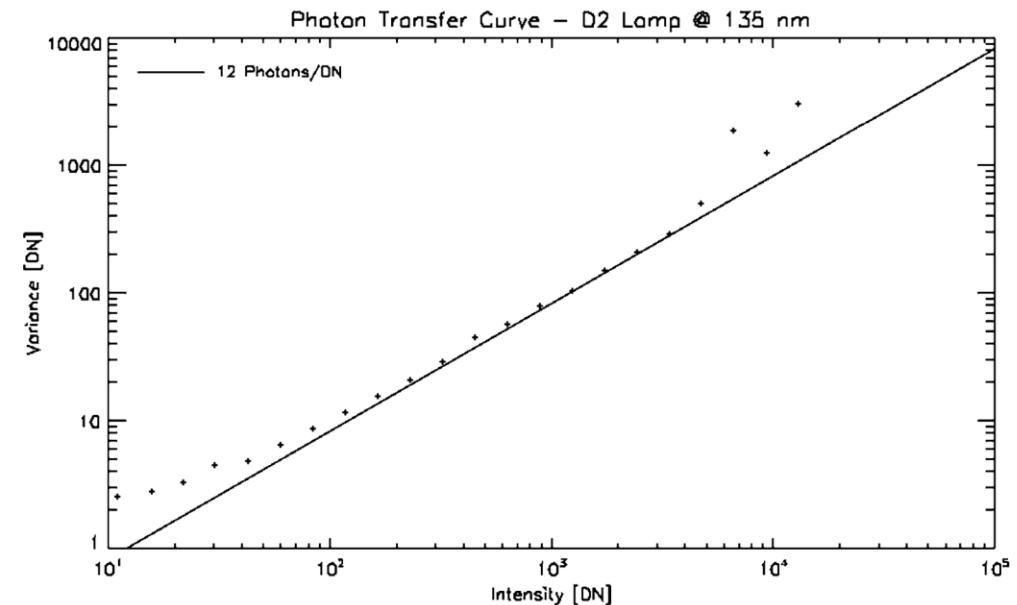


Figure 27 Photon transfer curve for the SJI CCD measured with a deuterium lamp at 1350 Å.

HST/WFC3 Noise

- Wide Field Camera 3 (WFC3) on Hubble also has this issue (Marinelli & Green 2024)
 - “Quantum yield” is lower
 - Expected ~ 1.7 @ 200 nm
 - Measured 1.09 @ 200 nm
 - SNR is $\sim 25\%$ better than expected

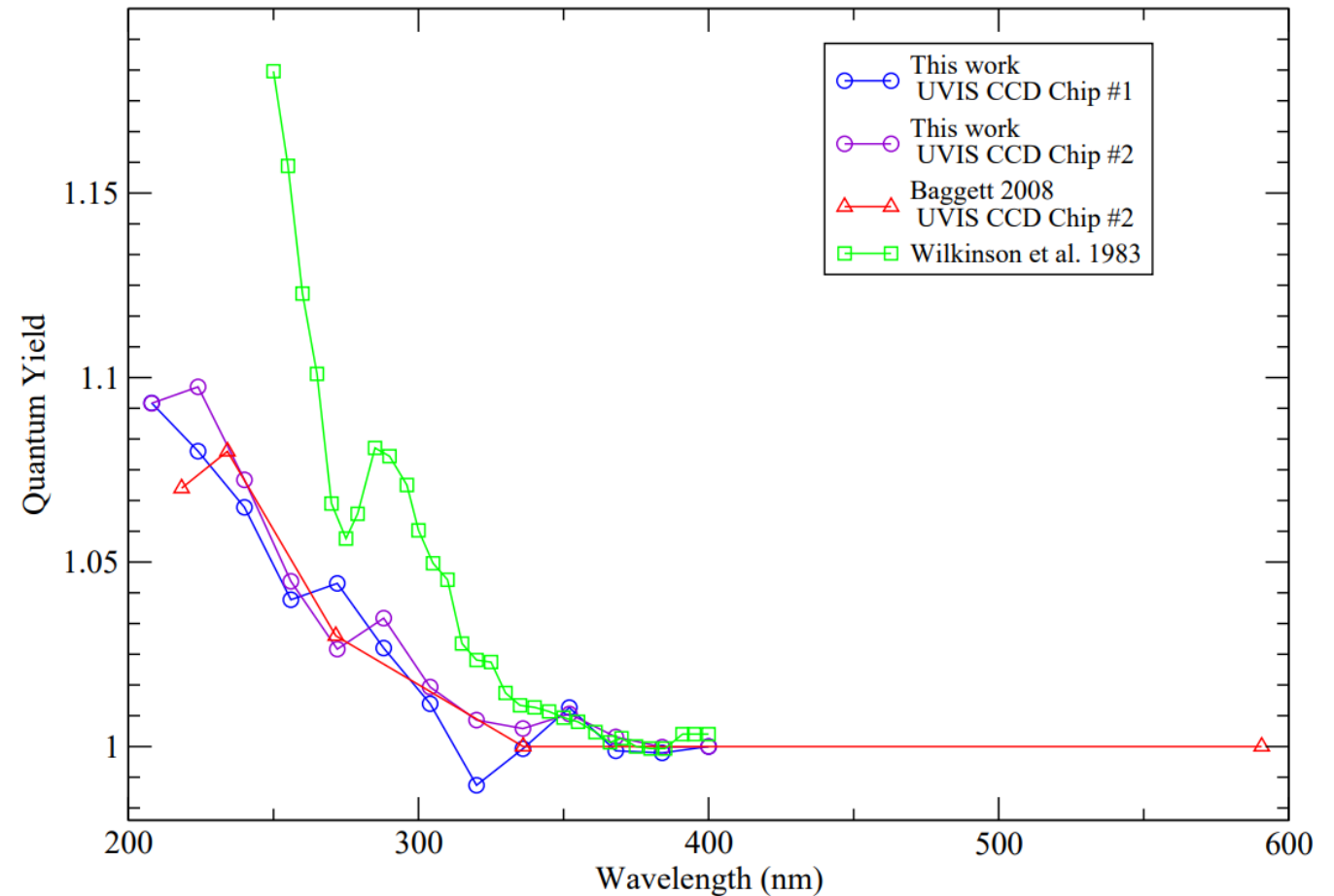


Figure 1. Our resulting quantum yields are plotted as a function of wavelength (blue and purple); the Baggett 2008 measurements are plotted in red and the quantum yield measurements of Si I by Wilkinson 1983 results are shown in green.

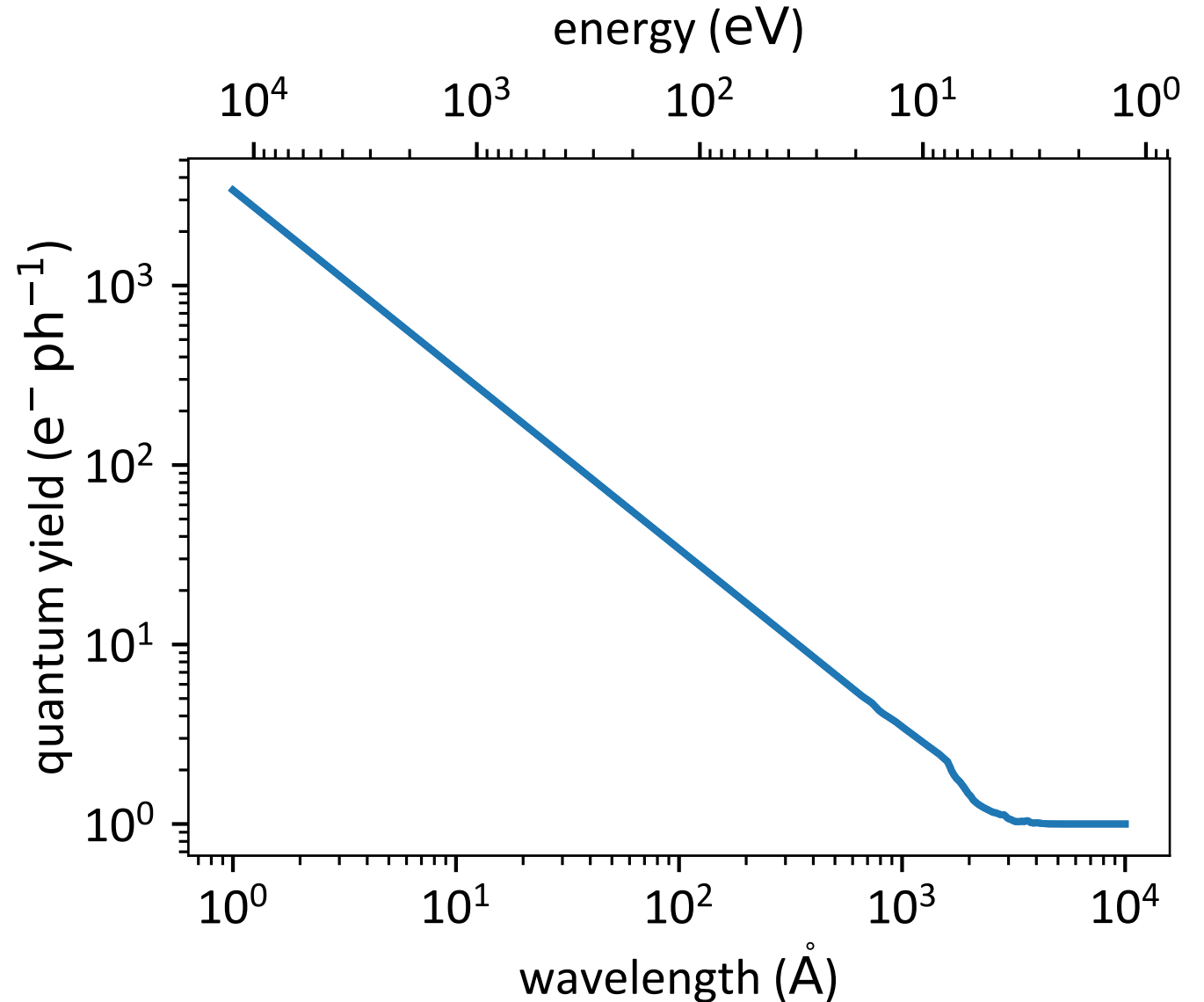
Outline

- Signal
 - Quantum yield
 - Charge-collection efficiency
 - Absorption
- Noise
 - Simple noise model
 - Shot noise
 - Fano noise
 - Partial-charge collection noise
- Results

Signal

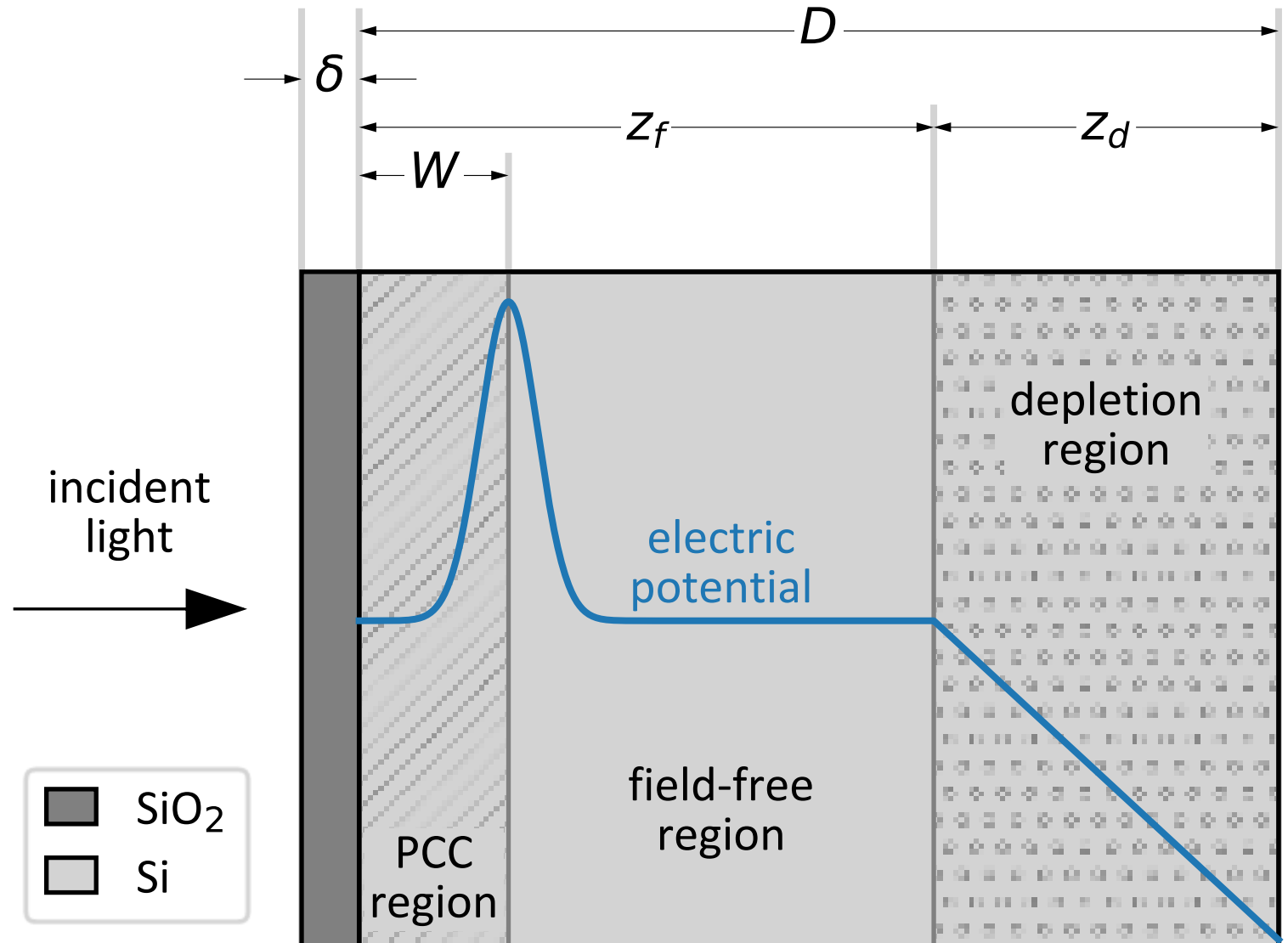
Quantum Yield

- Quantum yield (QY) is the number of electron-hole pairs generated per photon
- QY increases as photon energy increases
 - $QY = E / (3.65 \text{ eV})$ where E is the energy of the photon
- Ramanathan & Kurinsky (2020) QY model



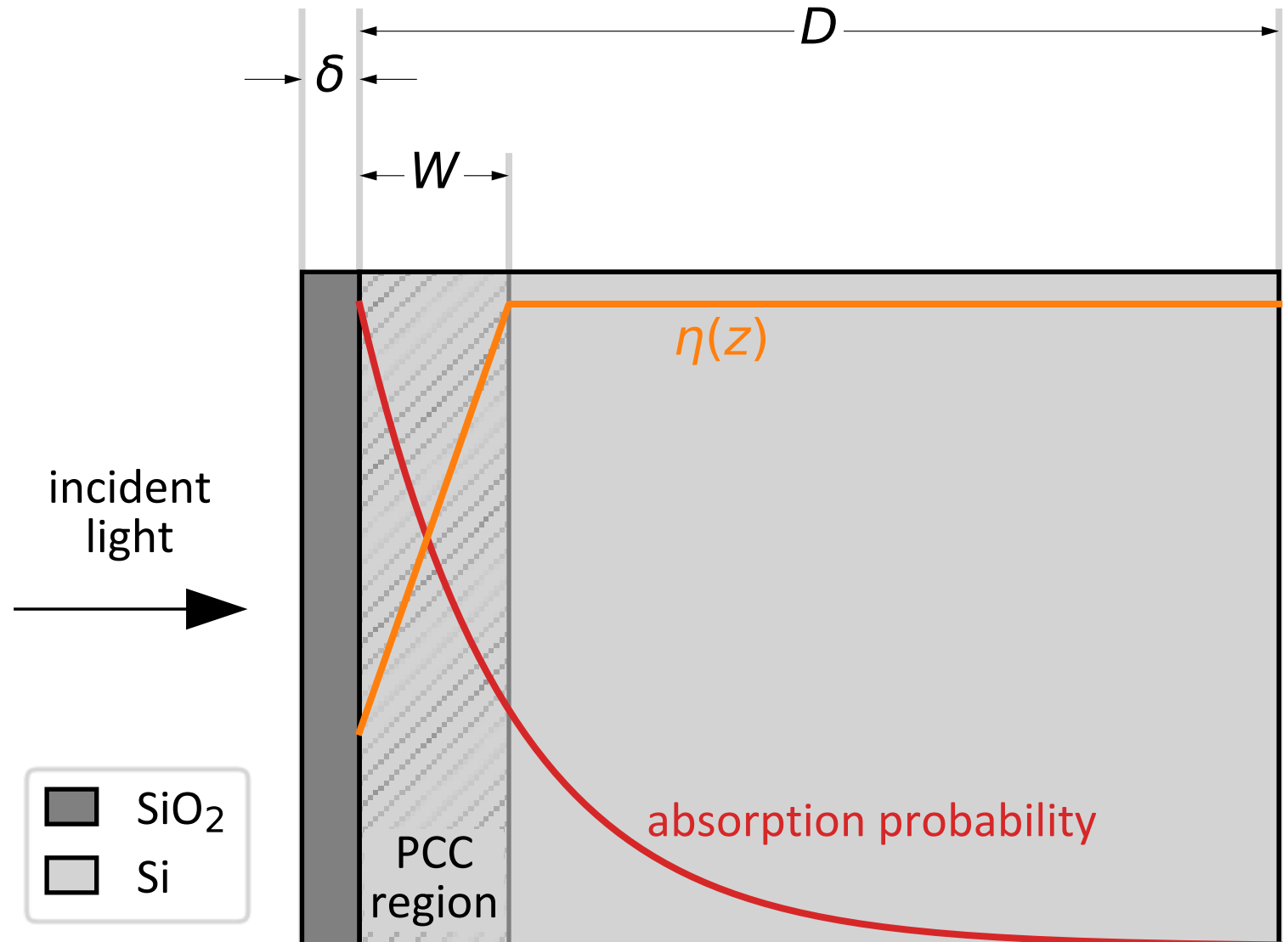
Back-illuminated Sensor Model

- Thin layer of silicon dioxide (~ 1 nm)
- Light-sensitive silicon ($16\text{ }\mu\text{m}$)
 - Partial-charge collection (PCC) region (~ 100 nm)
 - Field-free region
 - Depletion region ($8\text{ }\mu\text{m}$)



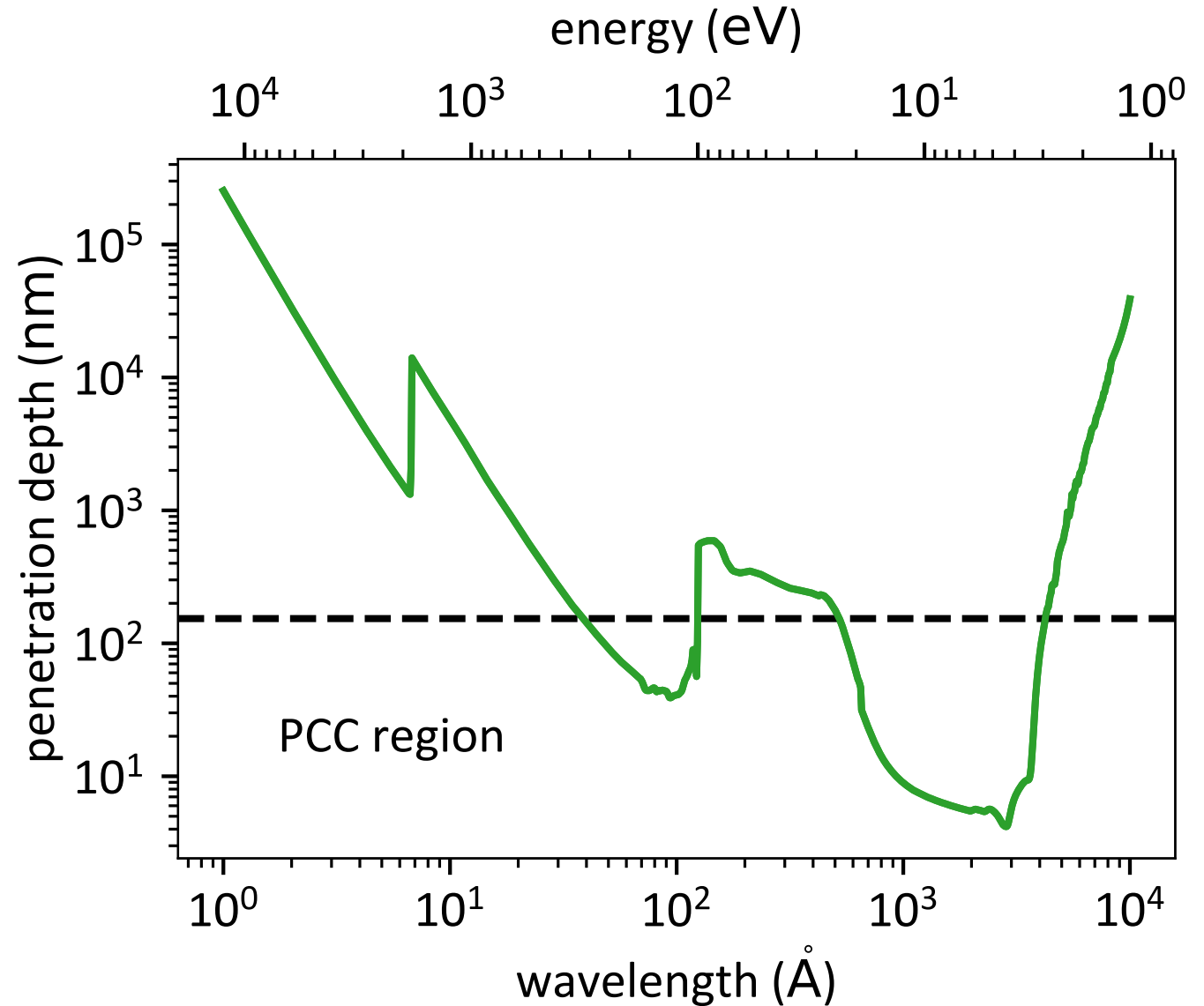
Partial-Charge Collection

- Photons are absorbed according to the Beer-Lambert law (red)
- Photons absorbed in PCC region lose some electrons
- Charge-collection efficiency (CCE) is the fraction of charge measured.
- Piecewise-linear differential CCE introduced by Stern et al. (1994)
 - Denoted $\eta(z)$



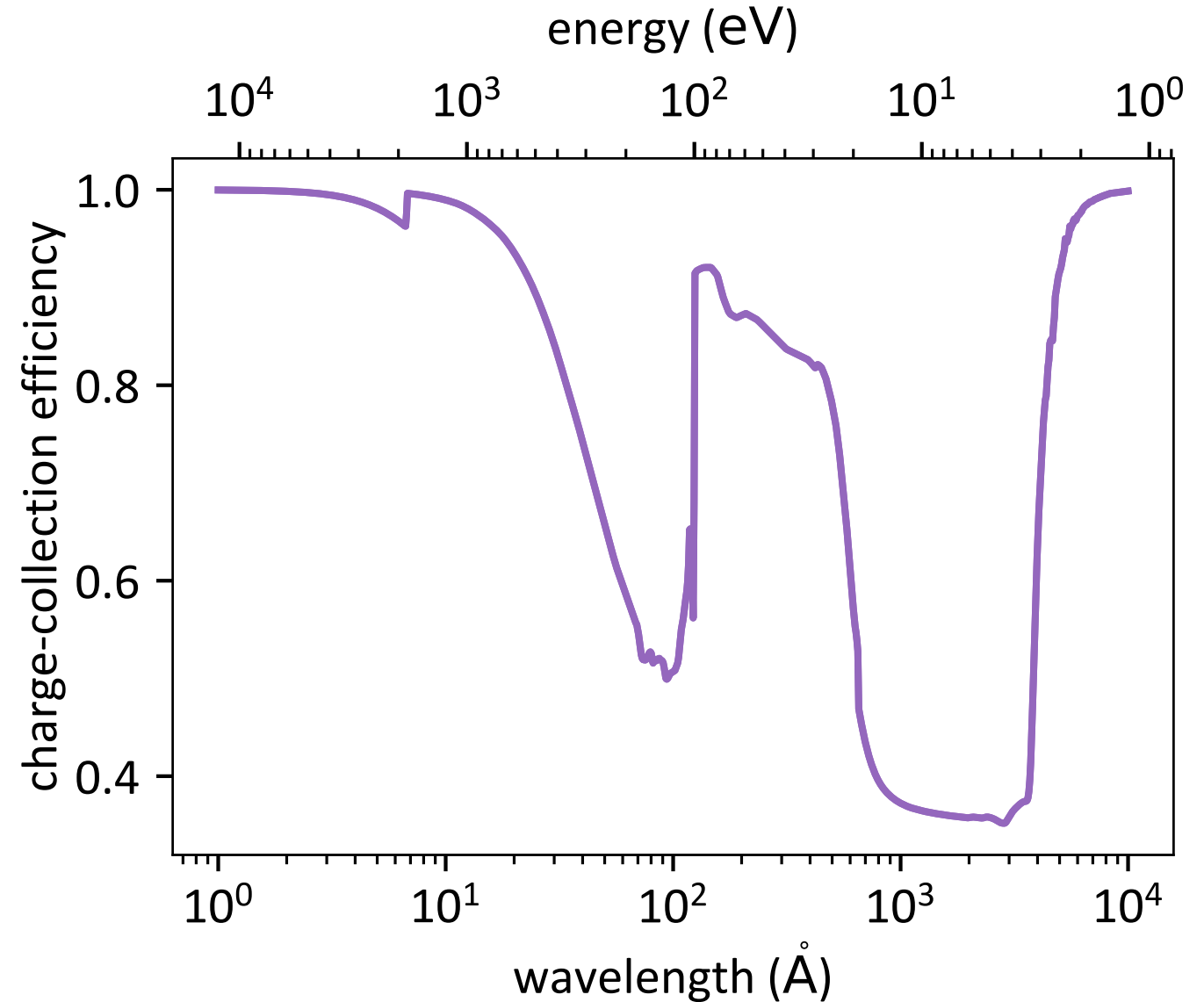
Penetration depth

- Some ultraviolet wavelengths have a penetration depth smaller than the PCC region
 - $\sim 40 - 100 \text{ \AA}$
 - $\sim 500 - 4500 \text{ \AA}$
- Sensor is much less efficient in these regions



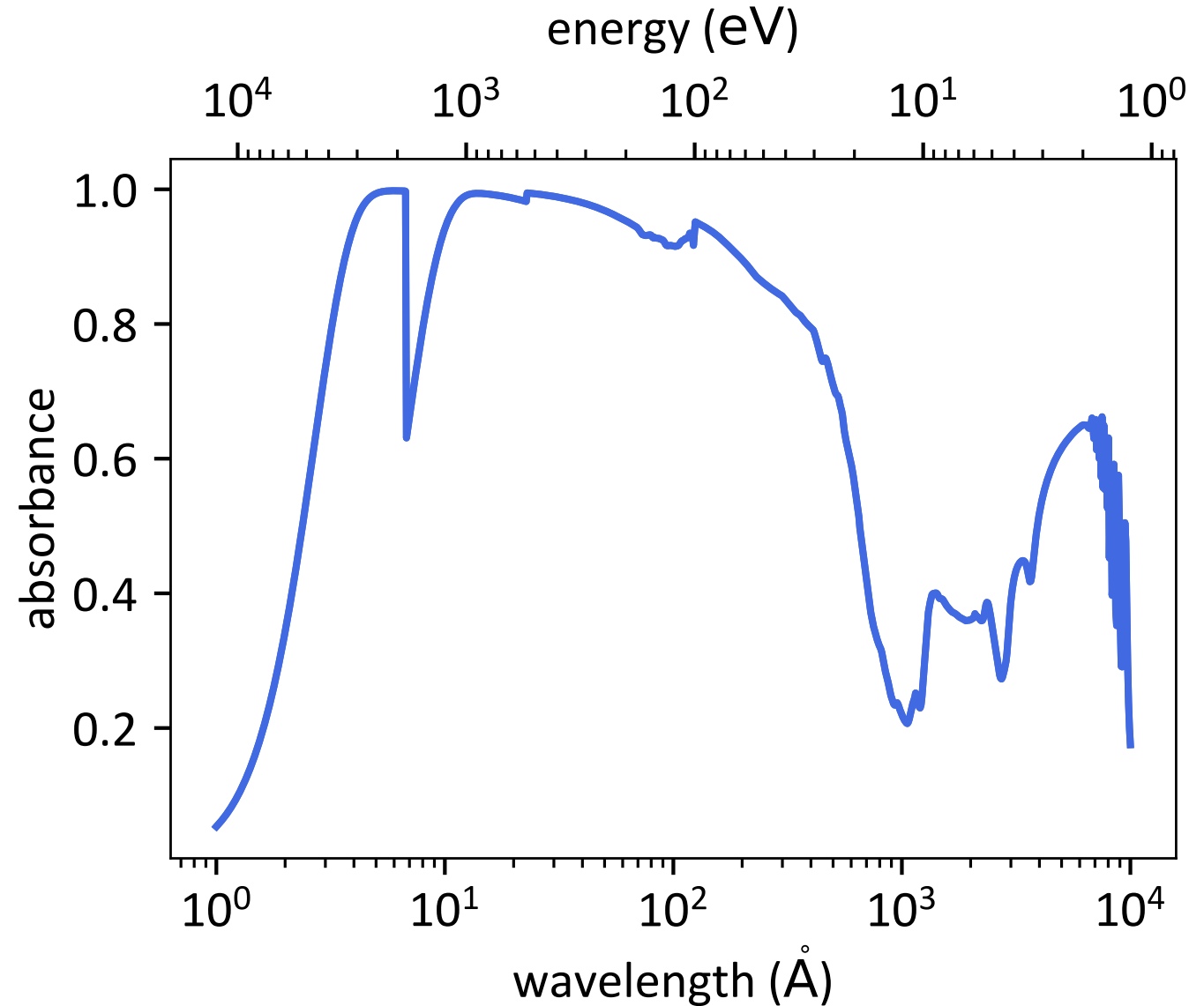
Average Charge-Collection Efficiency

- Integrate Beer-Lambert law against differential CCE
- Average CCE is needed for correct photometry



Absorbance

- Energy absorbed by light-sensitive layer
 - Reflections from each interface
 - Oxide absorption
 - Transmittance through entire sensor
- Computed using standard optical constants (Henke 1993, Palik 1997)



Quantum Efficiency

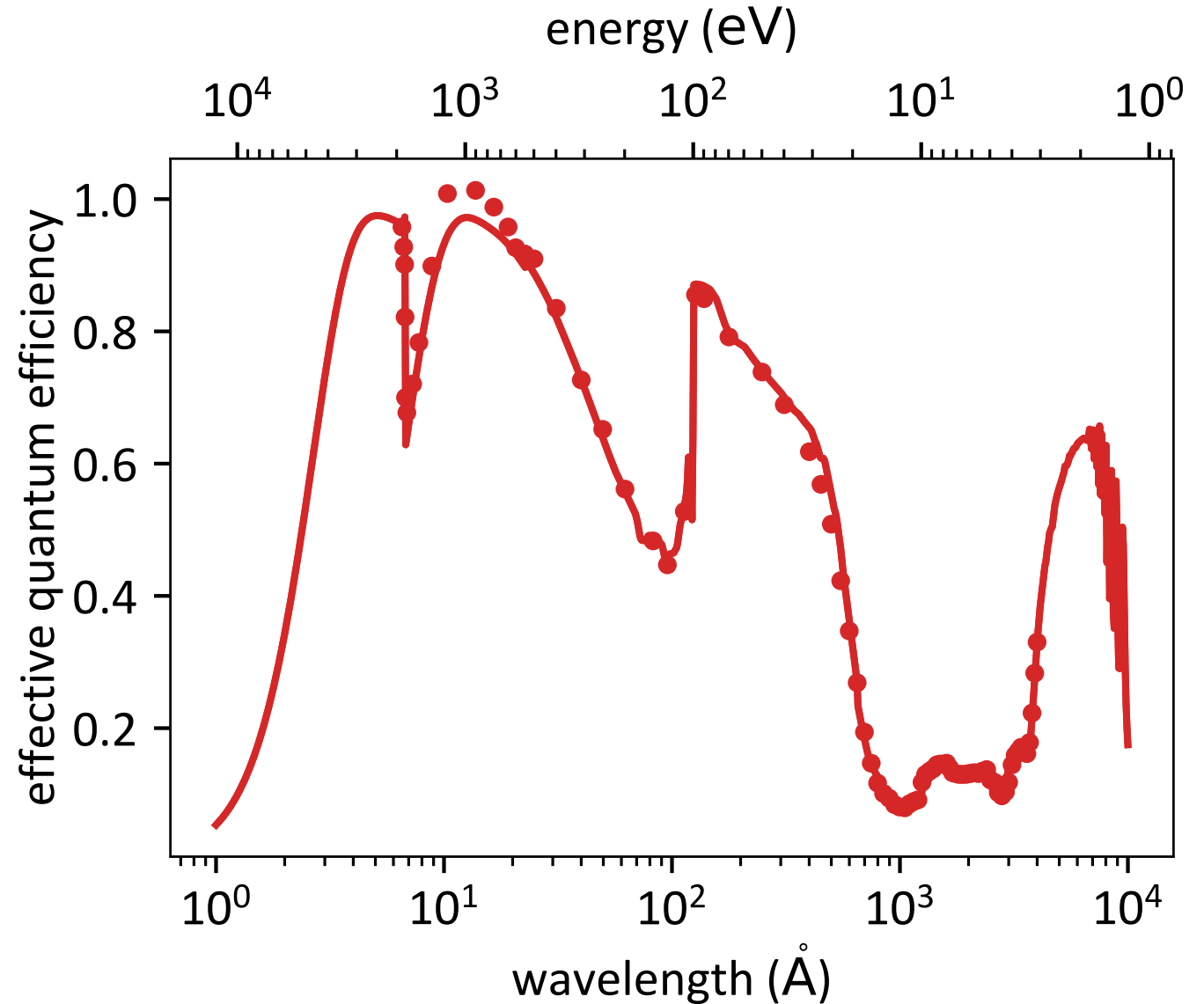
- Quantum Efficiency (QE) is the factor used to convert from incident photons to measured electrons
- Allowed to be larger than one!

$$\begin{aligned} \text{QE} &\equiv \frac{N_e}{N_\gamma} \\ &= \text{QY} \times \text{CCE} \times \text{Absorbance} \end{aligned}$$

Effective Quantum Efficiency (EQE)

$$\text{EQE} = \text{CCE} \times \text{Absorbance}$$

- Measured quantity
 - Compare sensor to NIST photodiode
- Between 0 and 1
- Often used to compute QE



Noise

Variance-to-mean ratio

$$\text{VMR}(X) = \frac{\text{Var}(X)}{\langle X \rangle}$$

- Convenient measure of noise
 - If X is Poisson-distributed, $\text{VMR}(X) = 1$
- Constant vs. signal, unlike SNR
- Equal to the slope of the photon-transfer curve
- Same units as X

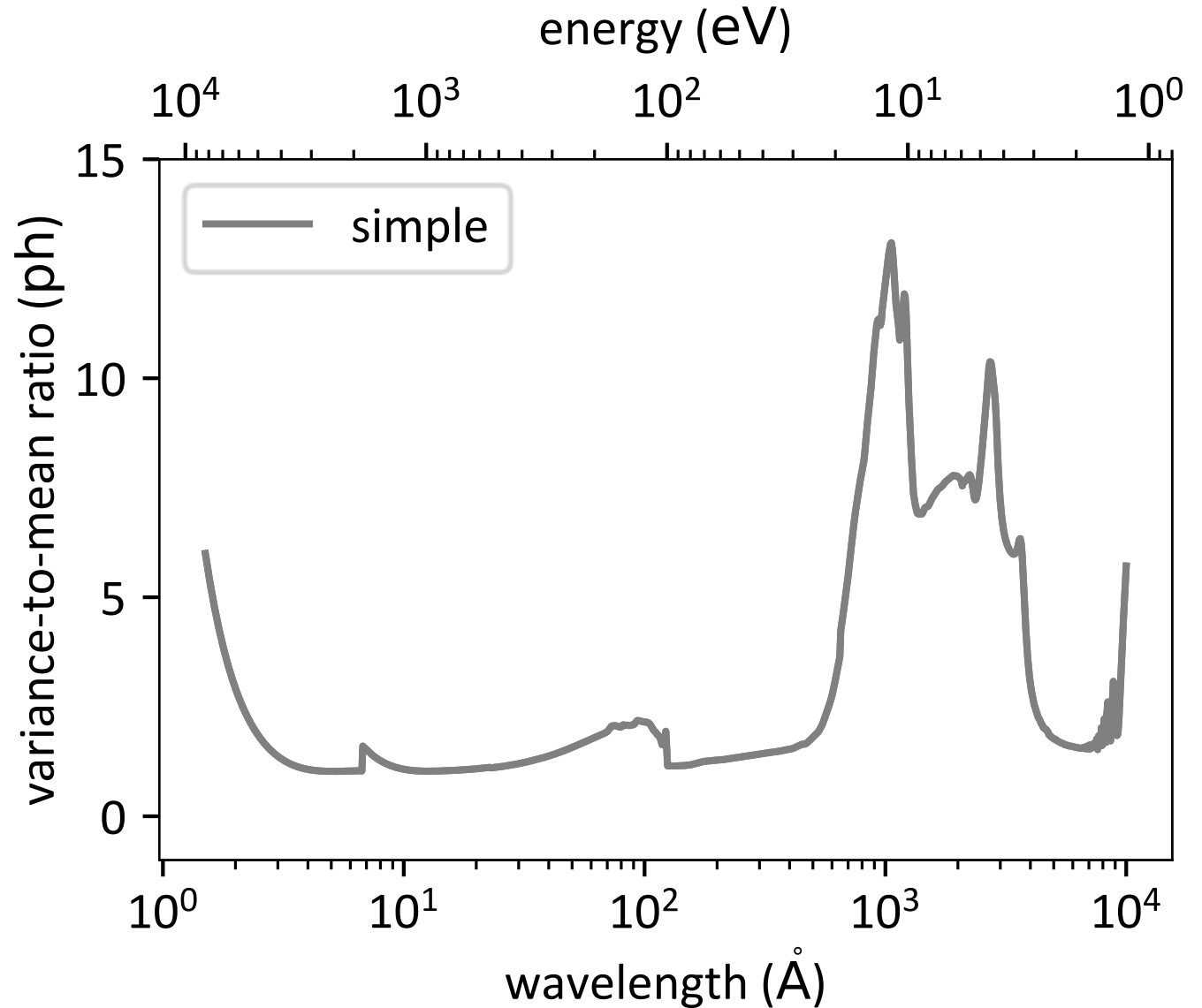
Simple Noise Model

- Shot noise only
- All-or-nothing charge collection
- Used by IRIS and WFC3

$$\sigma_{N_e} = \sqrt{\text{EQE} \times \langle N_\gamma \rangle}$$

$$\Rightarrow \text{VMR}(N_e) = \text{QY}$$

$$\Rightarrow \text{VMR}(N_\gamma) = \frac{\text{QY}}{\text{QE}} = \frac{1}{\text{EQE}}$$



A Slightly More Complicated Noise Model

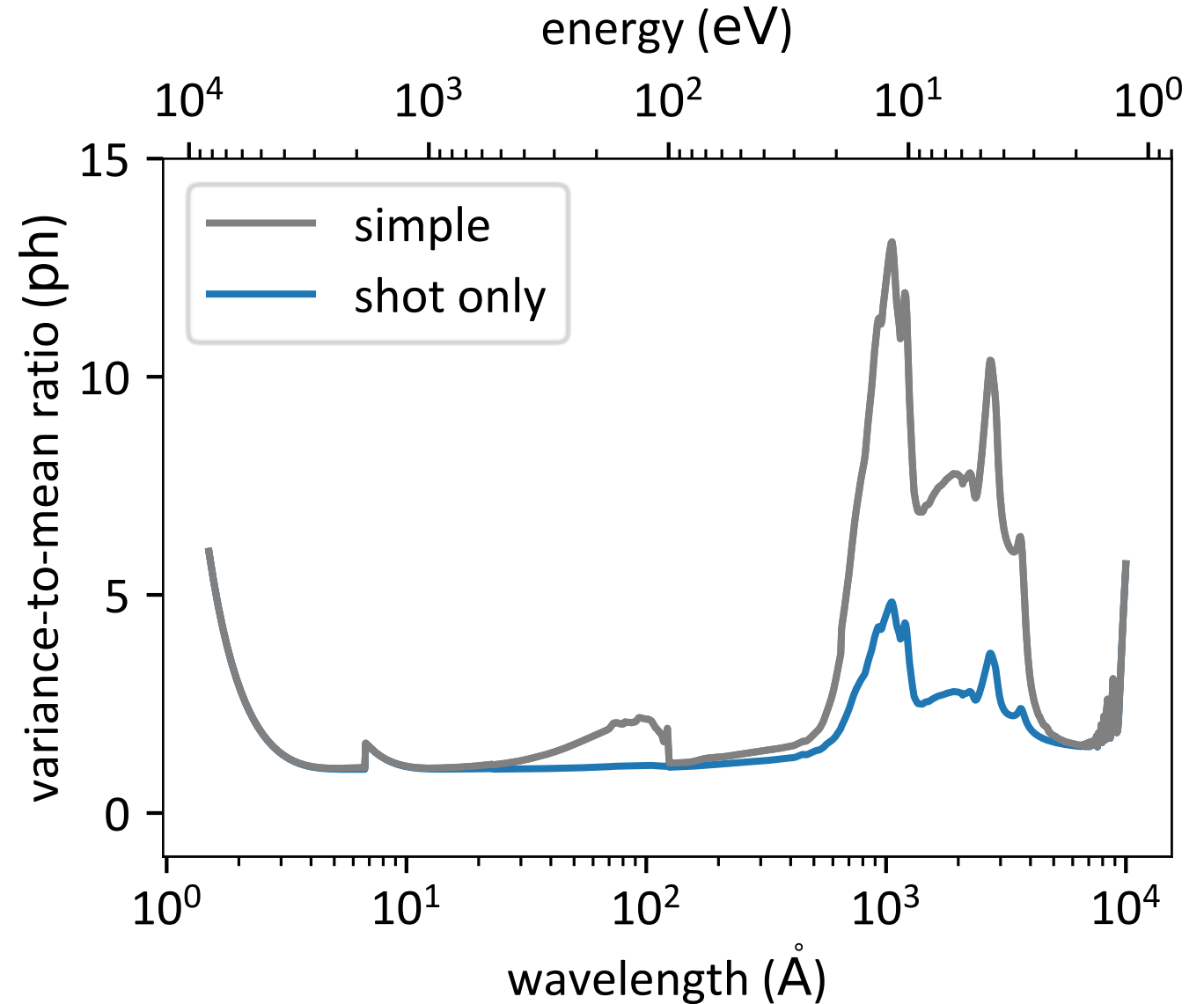
- Shot noise
- Fano noise
 - Inherent randomness in the charge-generation process
- **PCC noise**
 - Uncertain CCE due to random absorption depth

Shot Noise

- Absorbed photons are Poisson-distributed

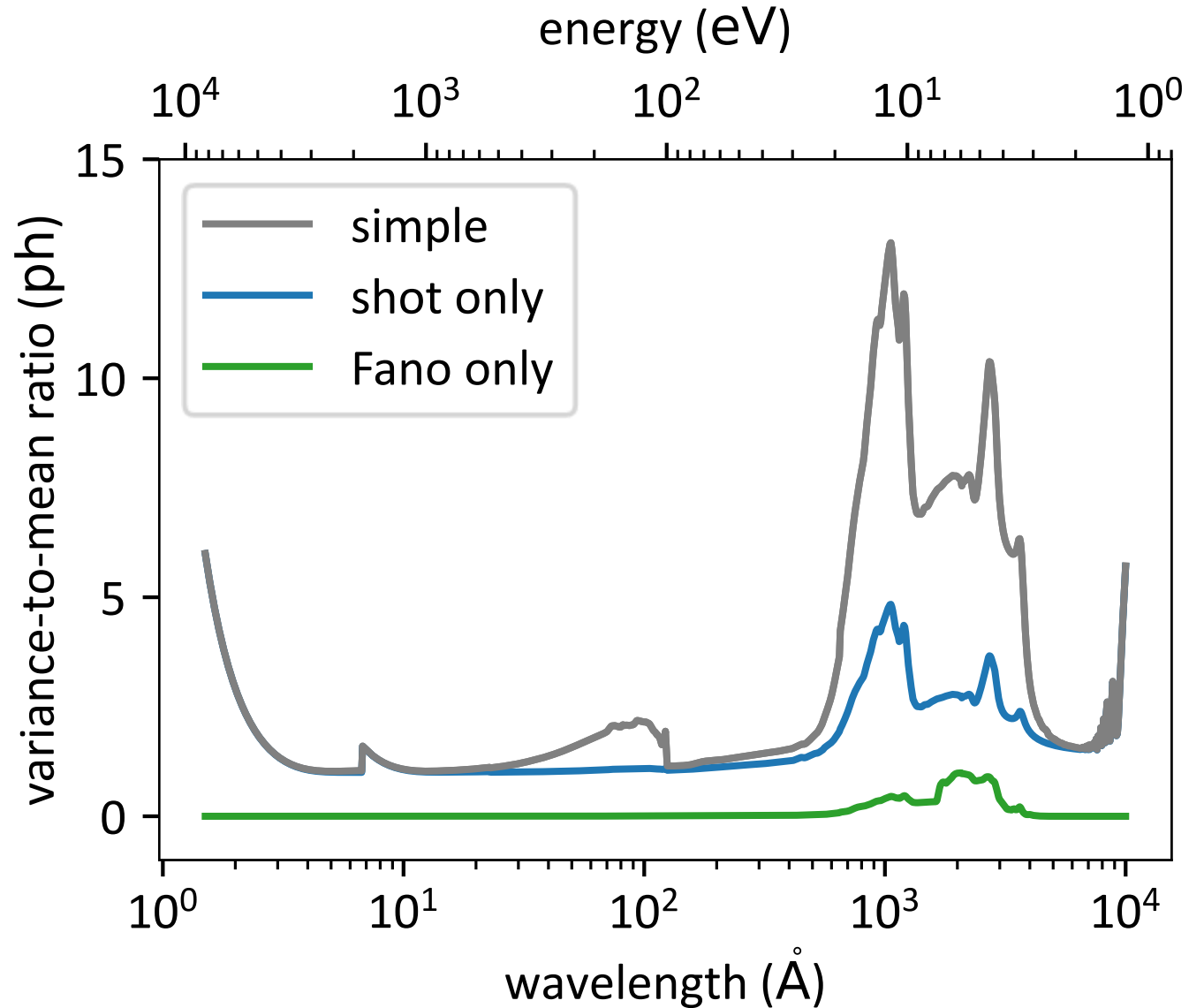
$$N'_\gamma \sim \text{Poisson}(A \times \langle N_\gamma \rangle)$$

$$\text{VMR}(N_\gamma) = \frac{1}{A}$$



Fano Noise

- Quantum yield uncertainty
- Described by Fano factor
 - $\mathcal{F} = \text{VMR}(\text{QY}) \simeq 0.1$
- Ramanathan & Kurinksy (2020) QY model
- Small compared to other noise sources

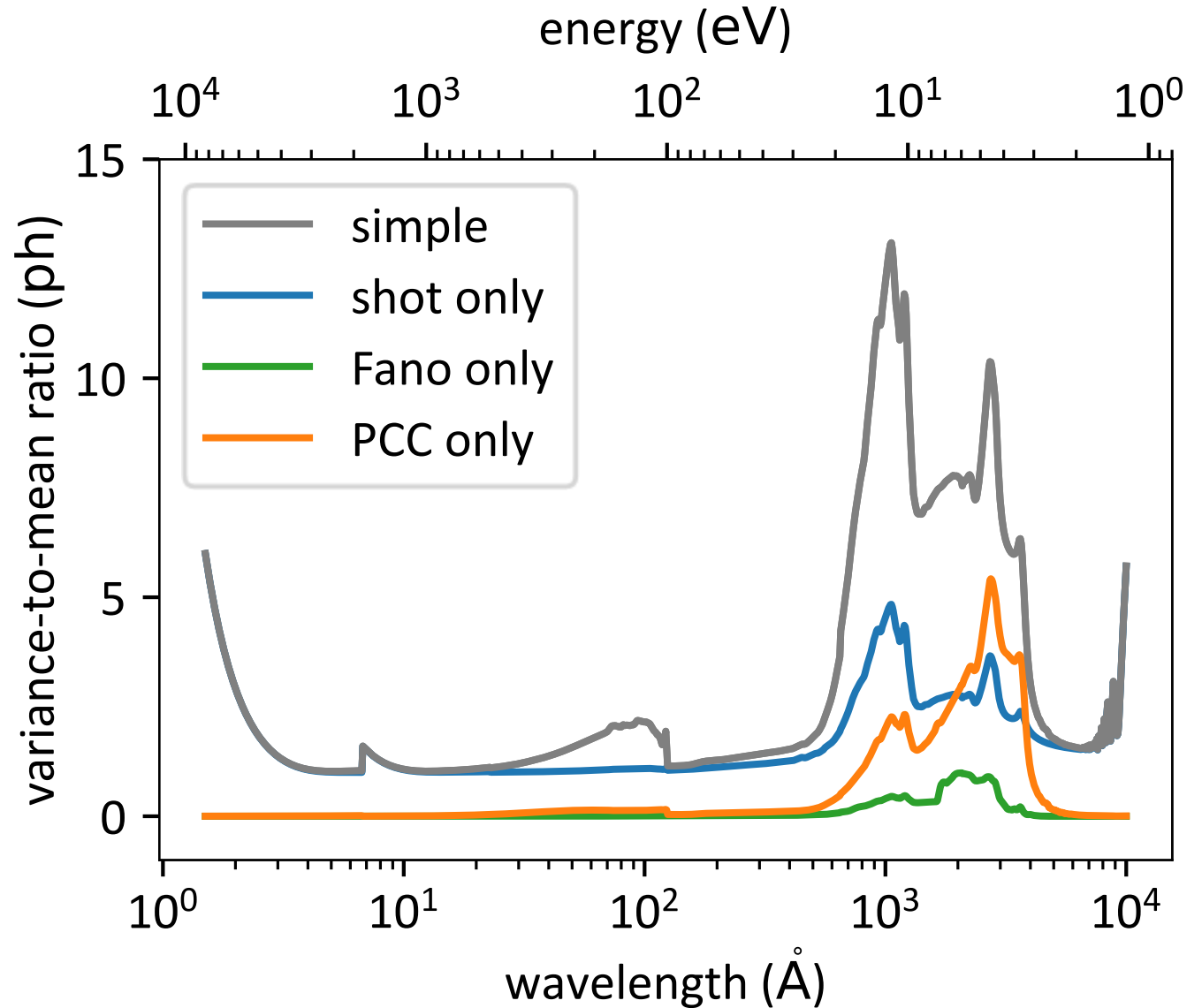


PCC Noise

- Since each photon is absorbed at a different depth, each experiences a different CCE.

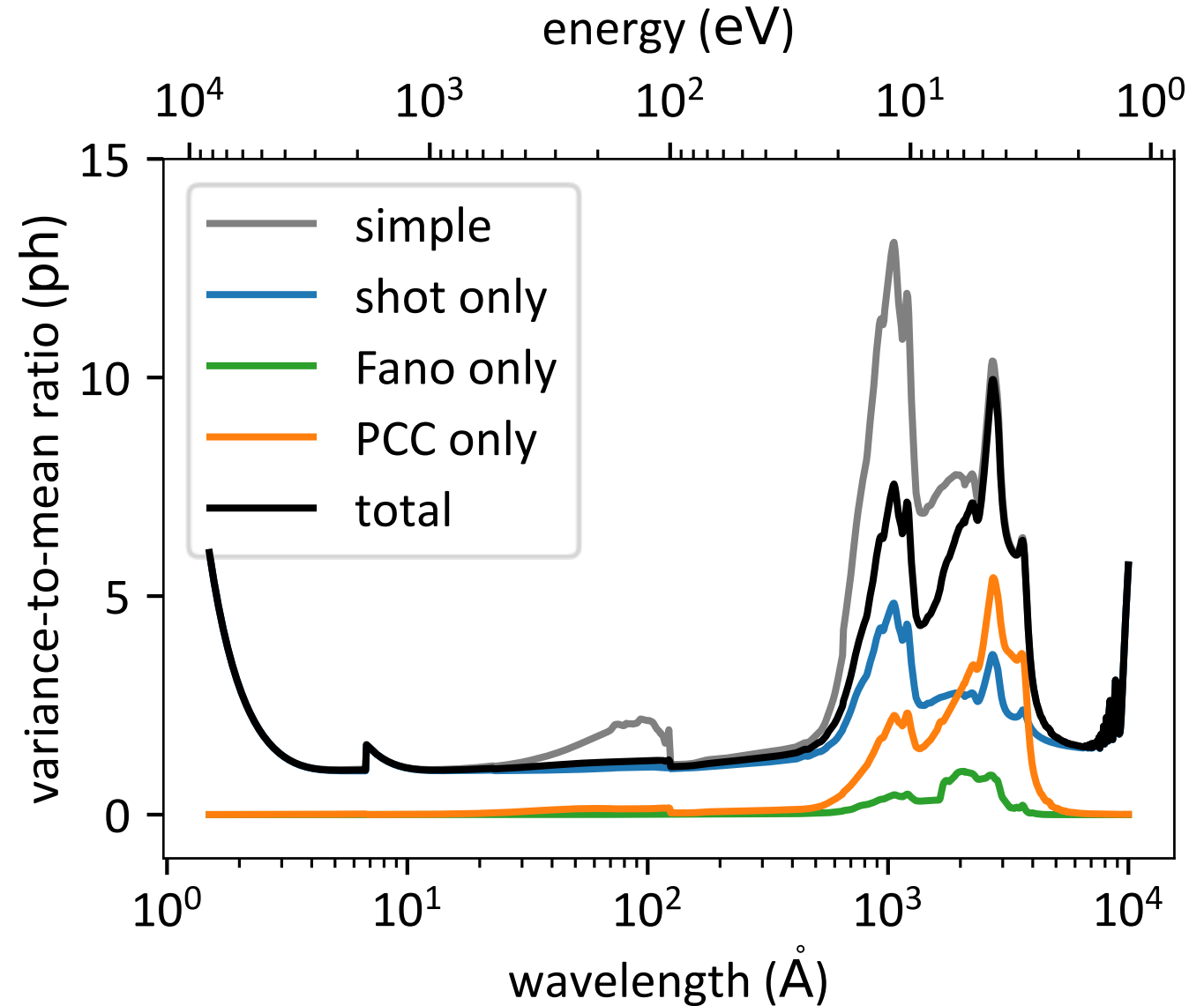
$$n_i \leftarrow \text{Binomial}(\text{QY}_i, \eta(z_i))$$

$$N_e = \sum_{i=0}^{N'_\gamma} n_i$$



Total Noise

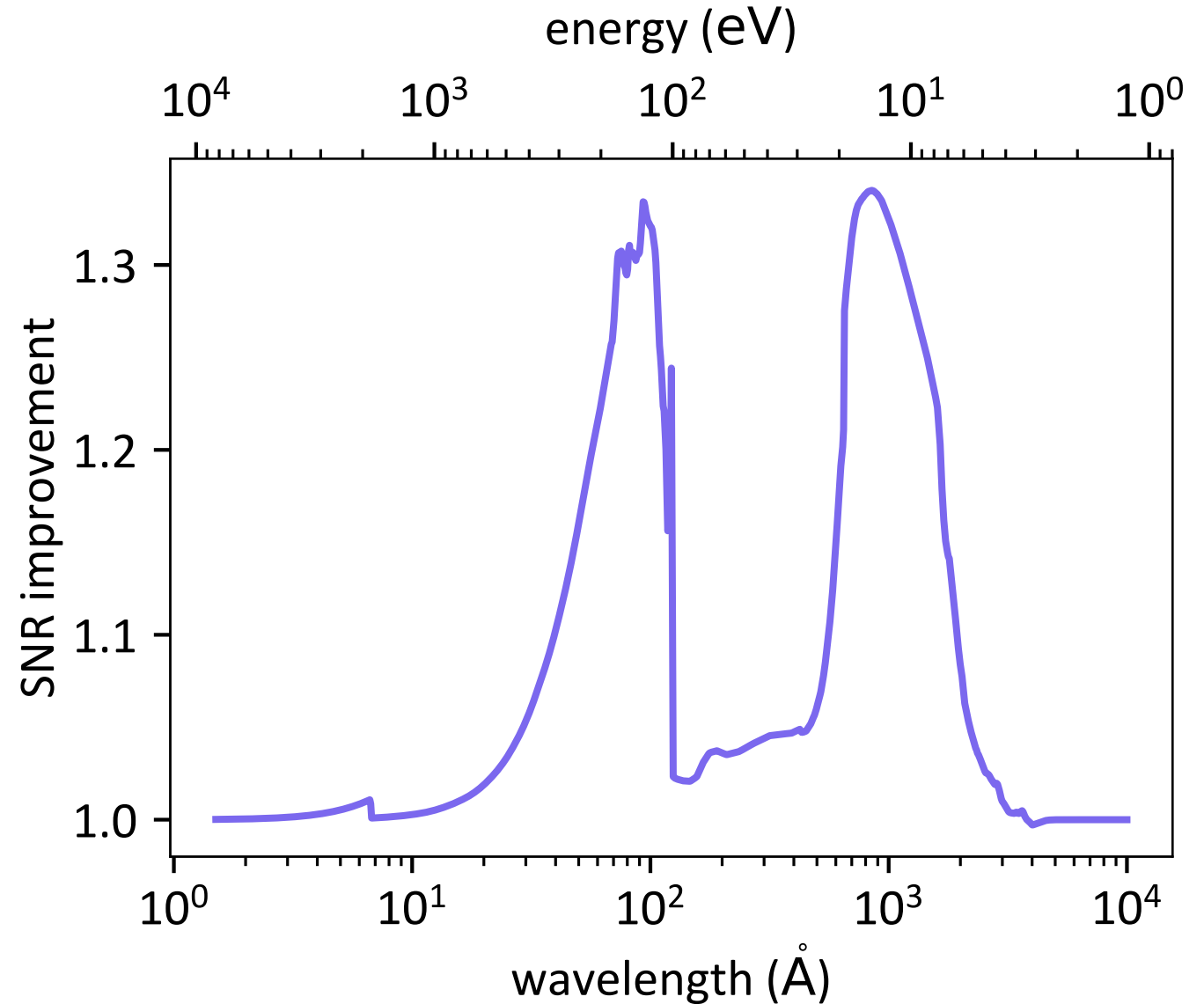
- Sum of shot, Fano, and PCC noise
- VMR is smaller than the simple model in the UV
- The partial events improve the noise performance



SNR improvement

$$= \sqrt{\frac{\text{simple VMR}}{\text{total VMR}}}$$

- SNR improves by up to ~30% compared to the simple model



Charge spreading?

- Charge spreading is when electrons diffuse into adjacent pixels.
- Proportional to size of field-free region
- Could explain the discrepancy since this increases the correlations between pixels

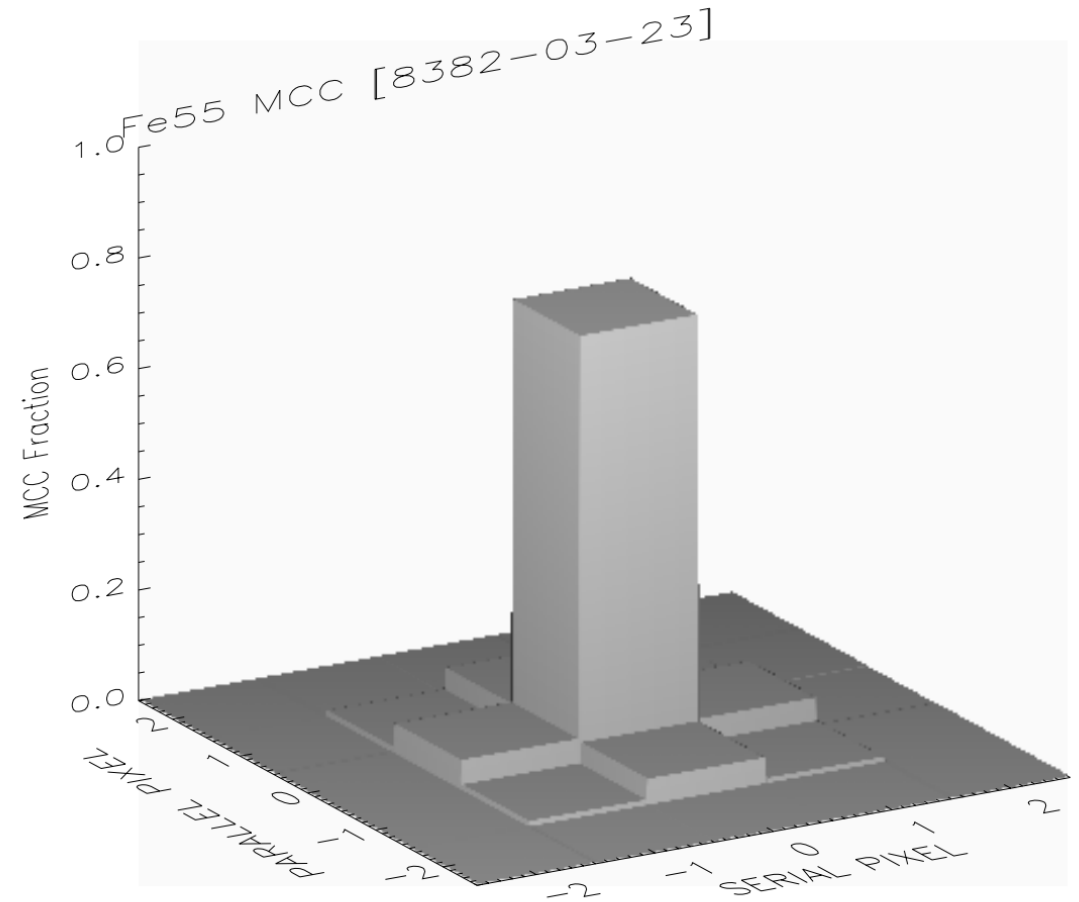
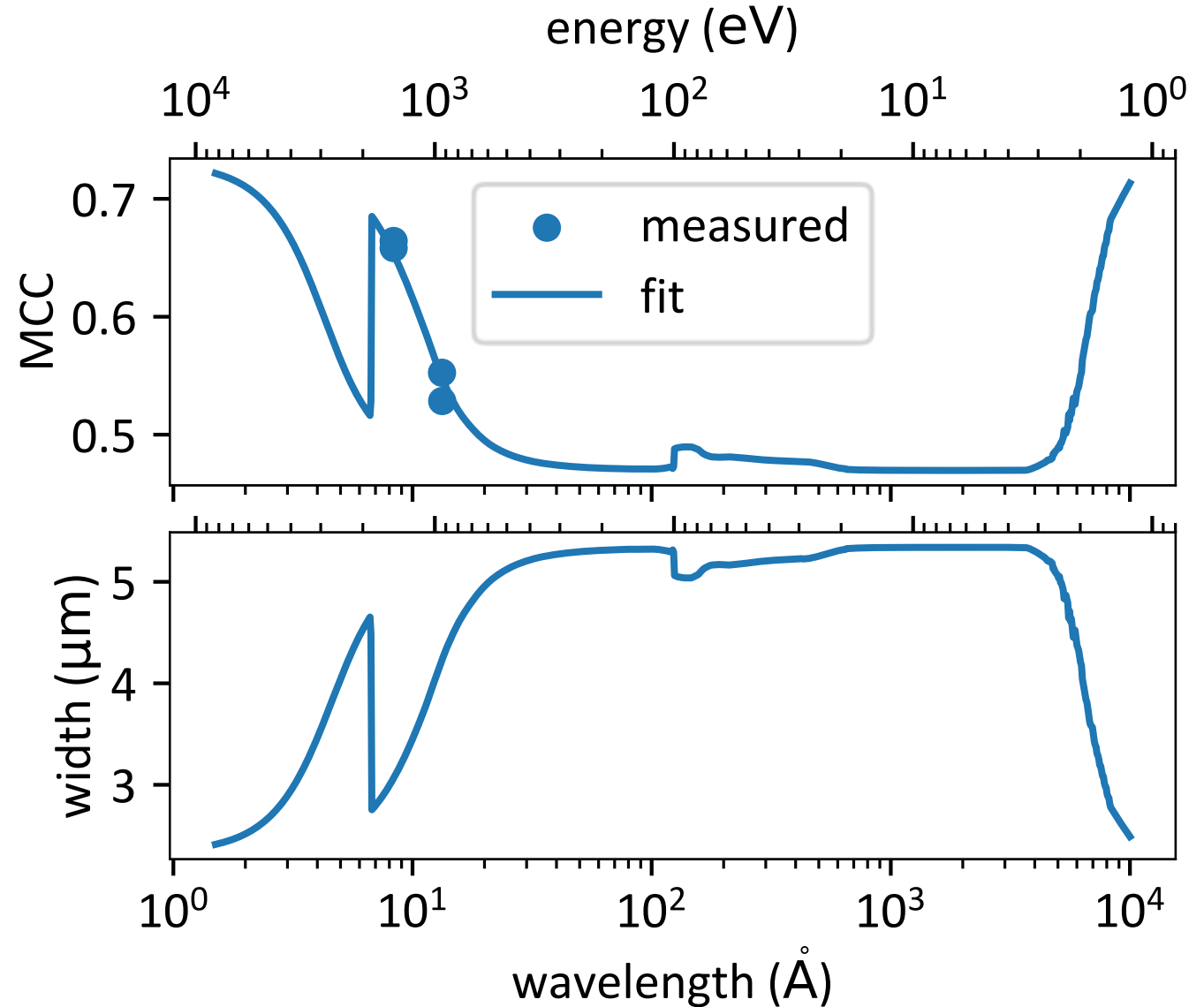


Figure 13. ^{55}Fe (5.9 keV) single photon PSF. MCC is 0.78 for central pixel.

Stern et al. (2003)

Mean Charge Capture (MCC)

- Average fraction of charge captured by central pixel
- Janesick (2001) charge spreading model
- Stern et al. (2004) measurements
- Fit size of field-free region
- Predicts a similar amount of charge spreading for both UV and visible light



Compare to Measurements

Instrument	λ_{UV} (Å)	λ_{vis} (Å)	model	measurement
IRIS	1350	4500	1.57	1.50
WFC3	2080	4000	1.14	1.09
	2240	4000	1.11	1.09
	2400	4000	1.08	1.07
	2560	4000	1.06	1.04
	2720	4000	1.04	1.04
	2880	4000	1.03	1.03
	3040	4000	1.01	1.01

Affected Instruments

- AIA
 - 94 Å
 - 1600 Å
 - 1700 Å
- IRIS
 - 1330 Å
 - 1400 Å
- MUSE
 - 108 Å

Instrument	λ (Å)	VMR (ph)	VMR (e ⁻)	improvement
AIA	94	1.23	20.14	1.34
	131	1.10	24.69	1.02
	171	1.16	18.44	1.03
	193	1.18	16.22	1.04
	211	1.20	14.88	1.04
	304	1.30	10.15	1.05
	335	1.33	9.20	1.05
	1600	4.92	1.47	1.23
	1700	5.55	1.38	1.18
IRIS	1330	4.41	1.65	1.28
	1400	4.36	1.60	1.27
	2796	9.68	1.07	1.03
	2832	9.37	1.07	1.03
MUSE	108	1.24	19.39	1.27
	171	1.16	18.42	1.04
	284	1.28	10.93	1.05

Conclusions

- Partial-charge collection is mostly responsible for IRIS noise mystery
- Charge spreading is less important
- Need to use a more complicated expression to compute noise
- The model predicts better SNR in some wavelengths than designers expected
- Results affect AIA, IRIS, WFC3, ESIS, FURST, and MUSE.
- Code provided in the `optika.sensors` Python module.
 - Published on PyPI